

Topics and sub topics of Chapter 1 Introduction

1. What Operating Systems Do
 - a. User View
 - b. System View
 - c. Defining OS
2. Computer-System Organization
 - a. Interrupts
 - b. Storage structure
 - c. I/O structure
3. Computer-System Architecture
 - a. Single Processor Operating System
 - b. Multi Processing Operating System
 - c. Clustered System
4. Operating-System Operations
 - a. Multiprogramming
 - b. Multi Processing
 - c. Dual Moed
 - d. Multi Mode
 - e. Timer
5. Resource Management
 - a. Process Management
 - b. Memory Management
 - c. File Management
 - d. Mass storage Management
 - e. Cache Management
 - f. I/O system Management
6. Security and Protection
7. Virtualization
8. Distributed Systems
9. Kernel Data Structures
 - a. List
 - b. Tree
 - c. Hash
 - d. bitmap
10. Computing Environments
 - a. Traditional Computing
 - b. Mobile Computing
 - c. Client Serve Computing
 - d. Peer to peer Computing
 - e. Cloud Computing
 - f. Real Time Embedded System
11. Free and Open-Source Operating

Topics and sub topics of Chapter 2 Operating System Structure

1. Operating System Services
2. User and operating system Interface
 - a. Command Line Interface
 - b. Graphical User Interface
 - c. Touch Screen Interface
 - d. Choice of interface
3. Systems
 - a. Example
 - b. Application Program Interface
 - c. Types of System calls
4. System Services
5. Linker and Loader
6. Operating System Design and Implementation
7. Operating System Structures
 - a. Monolithic
 - b. Layered
 - c. Modules
 - d. Microkernels
 - e. Hybrid
8. Building and booting operating system
9. Operating System Debugging
 - a. Failure Analysis
 - b. Performance Analysis
 - c. Tracing
 - d. BCC

Topics and sub topics of Chapter 3 Processes

1. Process Concept
 - a. Process
 - b. Process State
 - c. Process Control Block
 - d. Threads
2. Process Scheduling
 - a. Scheduling Queues
 - b. CPU scheduling
 - c. Context Switches
3. Operations on Processes
 - a. Process creation
 - b. Process Termination
4. Interprocess Communication
5. IPC shared memory system